

AI at Sea: From Maritime Hype to Operational Readiness

Why will Artificial Intelligence transform shipping only when data, people, systems and governance are ready to support it?

Abstract

Artificial Intelligence (AI) is now firmly part of the maritime transformation agenda. For an industry that carries much of global trade, the appeal is clear: safer operations, sharper decision-making, better fuel performance, stronger compliance, and more reliable sustainability outcomes. Yet the challenge is no longer whether AI has potential. It is whether maritime organizations are ready to turn that potential into trusted operational value. Across ship owners, operators, ports, regulators, engineers, and technology providers, expectations are high. But adoption remains uneven, often constrained by fragmented data, legacy systems, unclear problem statements, limited digital readiness, and a naturally cautious operating culture. This article examines the gap between ambition and execution and argues that maritime AI will succeed only when it is built on strong data foundations, clear use cases, domain expertise, responsible governance, and a realistic view of AI as a decision-support partner rather than a replacement for human judgment.



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Introduction

Artificial Intelligence is rapidly moving from conference discussions to practical maritime use cases. It promises better visibility, faster analysis, improved safety, and stronger environmental performance. But in marine operations, value does not come from adopting a tool alone. It comes from solving the right problem with reliable data, clear ownership, and people who understand both the technology and the realities of ship and shore operations. Shipping still operates through a complex mix of onboard machinery, shore-based systems, manual reporting, vendor platforms, and multiple stakeholders who often define the same data differently. This makes AI adoption a business transformation challenge as much as a technology challenge. Industry must therefore ask a more disciplined question: not simply what AI can do, **but whether the organization is ready to use it responsibly and on a scale.**

What Maritime Really Expects from AI

Shipping remains one of the most important pillars of the global economy, moving cargo across oceans and connecting producers, ports, and consumers. Yet it is also an industry where change has traditionally been measured, shaped by safety requirements, regulatory obligations, capital intensity, and operational risk. That makes the arrival of AI both exciting and challenging.

AI is no longer discussed only as a future technology. It is now part of boardroom conversations, fleet performance reviews, port modernization plans, and sustainability strategies. The question being asked across the sector is simple but important: **what practical role should AI play in maritime operations?**

From Automation to Intelligent Support

Although different stakeholders use different languages, their expectations are surprisingly aligned.

The industry does not simply want more automation. It wants systems that can interpret conditions, support judgment, and improve the quality and speed of decisions.

In practical terms, operators expect AI to function as an intelligent assistant that can:

- anticipate risks before they become incidents;
- recommend better decisions in near real time; and
- make sense of large, fragmented operational datasets.

Safety remains the first and most sensitive expectation. Whether in

navigation, machinery monitoring, cargo operations, or port movements, AI is expected to reduce uncertainty and highlight risk before it turns into an incident.

The second expectation is efficiency. With rising fuel costs, pressure on margins, and tighter environmental targets, organizations want AI to help them:

- Optimize routes
- Reduce fuel consumption
- Minimize downtime
- Improve fleet performance

Sustainability is another major driver. AI is expected to support better voyage planning, emissions monitoring, energy optimization, and evidence-based compliance with evolving regulatory requirements.

In short, maritime organizations are looking for AI that can observe, learn, predict, recommend, and improve operational confidence.

Where the Industry Stands Today

The current position, however, is more uneven than the ambition suggests.

AI adoption is increasing, but much of it is still cautious, isolated, and limited in scope.

Many organizations are experimenting with:

- Pilot projects
- Limited use cases
- Proof-of-concept initiatives

Yet only a smaller number have embedded AI deeply into daily operations or decision workflows.

Rather than a broad transformation, the industry is often seeing useful but narrow improvements.

AI is often used to:

- Automate repetitive workflows
- Support contract analysis
- Provide basic predictive insights

These applications are valuable, but they do not yet match the larger expectation of intelligent, connected, end-to-end maritime operations.

The most important signal is the level of trust.

Most maritime professionals are not yet comfortable allowing AI to make decisions independently.

They are more willing to use AI as a guide, warning system, analytical partner, or second opinion.

Why AI Is Not Yet Delivering at Scale

1. The Data Problem

Organizations today are collecting more information than ever and investing in stronger platforms. Still, many struggle to turn that information into action. Teams often spend considerable time locating, checking, cleaning, and reconciling data before it can be used. When confidence in the underlying data is weak, confidence in AI outputs naturally becomes weak as well.

AI depends on the quality, relevance, and reliability of the data behind it.

The real challenge is therefore not just data storage. It is data activation: making information discoverable, trusted, contextual, and available at the point where decisions are made.

In maritime environments, data is:

- Fragmented across systems
- Poorly structured
- Inconsistent between stakeholders

Without consistent, integrated, and well-governed data, even advanced AI tools produce uncertain or incomplete results.

If people cannot trust the data, they will not trust the recommendation generated from it.

What makes an enterprise AI-ready?

- clean, trustworthy, and contextual data;
- clear ownership of data, definitions, and governance;
- connected pipelines between AI models and live operational data;
- feedback loops that turn insight into action; and
- teams that can interpret and challenge AI recommendations.

Taken together, these issues show that many enterprises remain in an “insight only” phase. They can generate analysis, but they struggle to connect that analysis to execution, learning, and continuous improvement.

The business cost of poor data

- Productivity loss
- Increased costs
- Strategic missteps
- Revenue loss
- Customer churn
- Compliance risk

Recurring obstacles

- fragmented tools that make data stitching slow and error-prone;
- poor access, unclear permissions, and limited trust;
- missing definitions, context, and shared meaning;
- heavy IT dependency and delayed data availability.

Where better data creates the greatest impact

- faster and more confident decision-making;
- improved operational efficiency and profitability;
- stronger strategy, planning, and innovation;
- better customer experience and collaboration.



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2. Legacy Systems and Infrastructure

Vessels, ports, and maritime offices were not originally designed as fully connected digital ecosystems.

Many still operate on:

- Outdated hardware
- Non-integrated software systems
- Limited connectivity

Introducing AI into this environment cannot be treated as a simple software installation. It requires integration, modernization, connectivity, cybersecurity, governance, and a clear understanding of operational workflows.

3. Lack of Standardization

The maritime value chain includes ship managers, charterers, ports, classification bodies, regulators, logistics firms, vendors, and seafarers. Each may collect, label, store, and interpret data differently.

This lack of uniformity makes it difficult to deploy AI solutions across fleets, ports, and commercial networks on a scale.

AI thrives on integration, while maritime operations still often function through fragmented systems and stakeholder practices.

Technical standards such as ISO 19847 and ISO 19848 provide useful direction for shipboard data sharing and sensor-level structure. However, the wider commercial ecosystem remains fragmented. The challenge is not only defining data correctly, but also ensuring adoption across mixed fleets, older assets, vendors, and stakeholders with different priorities.

4. Skill Gaps and Workforce Readiness

The human dimension is one of the most important barriers to successful AI adoption.

The industry is undergoing a shift:

- From manual operations to digital workflows
- From experience-based decisions to data-driven insights

But the workforce is still catching up.

The capability gap lies between:

- Traditional maritime skills
- Emerging digital and analytical capabilities

The issue is not simply a shortage of manpower. It is a shortage of readiness, confidence, and shared language between maritime experts and digital teams.

Subject matter experts (Chief engineers, Masters, Fleet managers) are rarely embedded in design teams. Mariners need to be part of product design, not just end-users. AI outputs must be explainable, not black-box predictions. Without Subject Matter Experts' input, products miss the mark and fail to deliver.

5. Organizational Mindset and Resistance to Change

Maritime operations are naturally conservative because the consequences of failure can be severe.

Decisions are often conservative, shaped by:

- Safety concerns
- Regulatory pressures
- Established ways of working

AI challenges familiar ways of working and requires organizations to rethink how decisions are made, reviewed, and trusted. Even when technology exists, adoption is slowed down by:

- Resistance to change
- Lack of digital leadership
- Uncertainty about ROI

6. The Problem Statement Gap

One of the most common reasons AI initiatives lose direction is also one of the simplest:

Organizations do not always define clearly what problem AI is expected to solve.

Instead, they:

- Start with technology
- Experiment without a clear objective

- Expect transformation without redesigning processes

This leads to:

- Failed pilots
- Poor outcomes
- Frustration with AI initiatives

7. How AI Is Actually Being Used

Actual adoption is still relatively shallow in many organizations. AI is visible in analytics, but it is not yet deeply embedded in how information moves and how decisions are executed.

Many teams are experimenting at the interface level rather than redesigning the data pipelines, feedback loops, and operating processes that would make AI truly useful.

In many cases, AI remains a surface layer.

Natural language querying and dashboard assistance are often the easiest starting points, but the more transformative capabilities - automated insights, predictive analysis, recommendations, forecasting, and closed-loop decision support - require stronger data foundations and higher trust.

Until AI is trusted to improve data quality, support operational action, and learn from outcomes, it will remain more of a conversational interface than a transformation engine.

8. The Daily Reality for Data Consumers

Although dashboards are widely used, reporting remains fragmented in many companies. Dashboards, spreadsheets, manual summaries, and local extracts often exist side by side, producing different versions of the same metric.

Many teams still depend on 'Excel' files or manually prepared reports for leadership-facing KPIs. This creates delays, ownership confusion, version-control issues, and dependency on individual knowledge.

Where reporting methods are inconsistent, business intelligence adoption does not automatically create trust. Accuracy, consistency, lineage, and accountability remain fragile.

Common data platform frustrations

- unclear data ownership;
- too much clutter and too little documentation;
- unreliable, stale, or poorly governed data;
- time lost chasing permissions and lineage;
- unclear query costs or billing; and
- interfaces designed more for engineers than business users.

Platform promises that deserve caution

The most exaggerated platform promises are usually those that simplify human, organizational, and semantic complexity into neat marketing phrases. Users are rightly cautious of claims suggesting that everyone can build anything instantly, at any scale, with no trade-offs.

Common promises that should be treated with caution include:

- “Anyone can build or consume data products”;
- “Instant querying at any scale”;
- “Autonomous governance in one click”;
- “a single source of truth without operational discipline”;
- “real-time insights effortlessly”; and
- “Low-code transformation for everyone.”

The Shift Needed: From Hype to Execution

To gain real value from AI, the maritime industry must move beyond enthusiasm and begin focusing on disciplined execution.

The required shift is not only technological. It is strategic, operational, and cultural.

- From adopting tools → to building AI-driven strategies
- From isolated experiments → to scalable solutions
- From data silos → to integrated data ecosystems
- From automation → to intelligent augmentation
- From exclusion of SMEs → to inclusion in designing AI products

People must remain central to this transition. Organizations need to invest in:

- Upskilling workforce
- Building digital confidence
- Encouraging collaboration between domain experts and technologists

AI solutions in maritime must be continuously refined through input from those closest to operations. Feedback from crew, superintendents, fleet managers, port teams, and engineers is essential. Without this operational loop, even sophisticated systems can become disconnected from practical reality.

AI will not transform maritime operations through technical capability alone. It will succeed when it is responsibly embedded into how the industry observes, reasons, collaborates, decides, and learns.

The Future: A More Realistic View of AI

The future of maritime AI is often described through images of fully autonomous vessels and completely automated operations.

That future may arrive in stages, but it is unlikely to be the immediate reality for most of the industry.

A more practical near-term future will involve:

- AI assisting human decision-making
- Predictive systems preventing failures before they occur
- Connected ecosystems providing end-to-end visibility
- Human expertise augmented by intelligent systems

In this version of the future, AI is not a substitute for human expertise. It is a multiplier of human capability.

Conclusion: Readiness Before Reinvention

The maritime industry is not short of ambition. What it needs now is alignment between expectation and execution, technology and infrastructure, innovation and workforce readiness.

The key question is no longer only: ***“What can AI do for us?”***

It is also: ***“Are our data, people, processes, and governance ready for AI?”***

About the author

Venkat Krishna Soundarraja, Fellow of IME(I), blends 30 years of marine engineering and data science. Former Chief Engineer turned Chief Data Officer at Volteo Maritime Pte. Ltd., he drives digital transformation and sustainability in maritime operations. As educator, mentor, and researcher, he bridges theory with practice, translating complex data into clear insights while inspiring future engineers through teaching, writing, and innovation.

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